



Responsive student model in an intelligent tutoring system and its evaluation

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Abstract

In this paper, we introduce a new student responsive model to support students who use an Intelligent Tutoring System (ITS) as an E-Learning tool. We proposed a weighted-based model to estimate and suggest learning materials for students who are pursuing a computer-based course. We have built a brand new ITS called WinITS with our proposed responsive student model and deployed it in Hanoi National University of Education-Vietnam (HNUE) in the second semester of the school year 2019-2020 with a computer science course. To compare the effectiveness of applying ITS to the students, we compare test results and analyze some other aspects related to the course. On the other hand, we conducted a survey between two groups: with and without using WinITS. 63 students are volunteers who participated in the case study. Before learning, 43 students from Group 1 will take a short survey of the Felder-Silverman questionnaire to identify learning styles, after that, they go through all the lessons from the course under the support of WinITS, the lessons will be chosen to satisfy student's need. On another side, 18 students from Group 2 will make the same test to compare the result to Group 1. In the range of research, we illustrate that our implementation shows some encouraging results such as reducing learning time, improving test score by 1.13 standard deviations, and making the lesson more interesting and flexible. The results have revealed some advantages of studying with computer-added compared to the traditional class in various ways and showed the effectiveness of the proposed model in Intelligent Tutoring Systems.

Keywords Learning styles · Intelligent tutoring systems · Adaptive learning · Responsive student model

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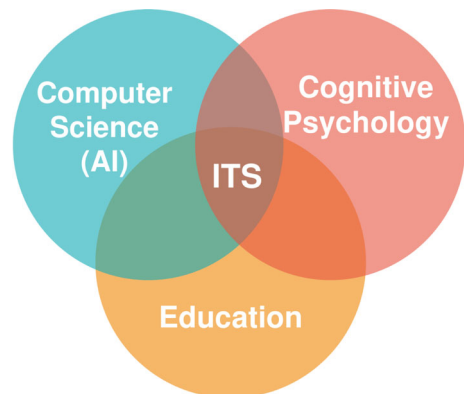
Extended author information available on the last page of the article.

1 Introduction

New technologies have taken a huge role in education and make significant improvements in this field day by day. Their application not only reduces the effective cost in the pedagogical process but also changes the way we learn. In education, technologies are changing vicinity, resources, and even methodology and it creates opportunities to support many useful methods of teaching (Georgieva et al. 2003). Thus, it is necessary to build a systematic which called Intelligent Tutoring System. It can be described as a system that can adapt to each learner. According to Oxman and Wong (2014), adaptive learning refers broadly to a learning process where the content taught or the way such content is presented changes, or “adapts,” based on the responses of the individual student. Since the early 1970s, the combination of Artificial Intelligence, Cognitive Science, and Education (Fig. 1) to create intelligent agents that can act as tutors in computer-aided-instruction (CAI) (Carbonell 1970). Generative CAI was the main precursor of ITSs which was introduced in the 1980s with the addition of the letter I, which stand for Intelligent (Sleeman and Brown 1982).

Recently, Artificial Intelligence (AI) is widely applied to ITS for adaptive purposes or classification/clustering process, especially in the field of action recognition and shows state-of-the-art results. AI supports ITS in tracking student progress, providing next-step hints and automatically finding problems of learners and aiding them to solve. Besides that, some researches show that learning objects have a direct impact on the selection of learning style (Graf and Liu 2010; El-Bishouty et al. 2018). According to that promising achievement, our work is focused on building a model that combines learning style, learning model which are understood as education part, user’s attention level as cognitive psychology, and student’s ability estimation which used machine learning (AI) to create a pathway to finish the computer-based course with the shortest time and highest performance. In the field of learning style, we deploy Felder-Silverman learning style model to evaluate the learning characteristics of learners. We also classify student actions in learning time for identifying interesting levels. Four kinds of learning materials are chosen to fit students’ behaviors. Based on the students’ characteristics which are analyzed carefully to map it into the

Fig. 1 The components of an Intelligent Tutoring System from Nwana (1990)



system resource, our ITS will choose the suitable materials for each student. Pedagogy and intelligence are being experimented with to incorporate into the ITS. After finish learning, students who take part in this ITS will join into a test to compare with other groups to illustrate the benefit of using ITS. Our survey shows a significant improvement between the two groups in many aspects. This result encourages us to go further with new improvements to better support learners in the manner of teaching innovation movement.

In this paper, we present our work related to building a new ITS based on Windows platform. This interactive e-learning tool allows customizing the documentation for specific learning styles. It itself can change to adapt to the student's requirements based on the characteristics and behaviors of the learner. Felder and Silverman learning style model and machine learning methods were applied to help students to improve their performance in a computer-based course.

The paper is organized as follows. Firstly, we represent previous work (Section 2) related to the paper included recent researches, used models, developed ITS and their size of implementation. Secondly, we discuss about our method and models to build up our system such as learning style models, learning models, besides that, we propose our responsive student model in the Section 3. This is followed by the motivation for the experiment in Section 4. In this section, we introduce our experiment by constructing a brand new Intelligent Tutoring System called **WinITS** which applied the learning style, learning model and responsive student above. The results are subsequently presented in the same section. Section 5 will discuss about the results in the context of similar studies in the area, after which the paper is concluded with future directions in Section 6.

2 Related work

In recent years, there are many innovative ITS have been built and deployed in schools. Together with the development of computer science, especially Artificial Intelligent, new AI ITSs have been shown to be effective in supporting teaching certain subjects. Two main objects of this application are school students and university students. According to the review of Mousavinasab et al. (2018), 37.73% of recent ITSs are in the computer science domain and 75% of the learner are university students. We summarized several previous researches and their size of implementation in Table 1.

Wei and Blank (2006, 2007) used Dynamic Bayesian Networks to build student models with some features such as students' performance history, prerequisite relationships, concept to solution step relationships, and student real-time responsiveness to help students learn, step by step, an object-oriented design course. They used CIMEL-ITS (Moritz et al. 2005) as an interactive system that provides one-on-one tutoring to help beginners in a CS1 course to study the design-first approach.

In the field of integration of learning style and learning resources, we investigate the relationship between Felder-Silverman learning style with the lesson's documentation. Savic and Konjovic (2009) built an e-learning system based on the Sakai open-source learning management system (Source 2020). They used the

Table 1 Previous works using ITS/LMS to adapt student in computer-based courses

Authors	Learning style model	Number of students	ITS/LMS
Fang Wei	various	49	CIMEL-ITS
Graf and Liu & Kinshuk	Felder & Silverman	75	Moodle
Diana C Muñoz et al.	Honey-Alonso	40	SIAS-ITS
Annabel Latham et al.	Felder & Silverman	63	Oscar CITS
Yang et al.	Felder & Silverman	54	AMDPC
Kai Ming LI	Felder & Silverman	39	Wikis
El-Bishouty et al.	Felder & Silverman	25	Moodle
Our work	Felder & Silverman	61	WinITS

Felder-Silverman model to identify student style and built related resources such as *assessment, presentation, example, illustration, summary* to adapt to students.

Franzoni et al. (2008) built Adaptive taxonomies for four learning styles dimensions to create a relationship between Electronic Media (EM) with Teaching Strategy (TS) through the Learning Style Dimension (LSD). These help students select the type of teaching strategies and the most appropriate electronic media.

Latham et al. (2014) built their own system called Oscar Conversational ITS (Latham et al. 2010) to allow students to construct their knowledge through discussion. The system selected the best fitting adaptation per question independently of the tutorial domain and the learning styles model (Felder-Silverman) adopted. The pilot was implemented on 63 students and the test scores before and after the pilot were compared.

Muñoz et al. (2010) applied the Honey-Alonso CHAEA learning style model (Alonso et al. 2007) and used SIAS-ITS as an adaptive system. The experiment was conducted on 40 medical students who totally are not familiar with the virtual learning environment. The participants were split into 2 groups. The first group included 22 students who tried to study on SIAS-ITS, the second one is 22 participants who used Moodle as a Learning Management System (LMS) (Moodle 2020). Their results show the efficiency and systems' usability were improved using the Intelligent Tutoring System.

Graf and Liu (2010) focused on the context of the learning object/activity and investigated the sequence of kinds of learning objects/activities in which students go through the course. They confirmed that the students' navigational behavior depending on their learning styles. This discovery can be seen as a basic requirement for providing adaptivity with respect to learning styles.

Yang et al. (2013) have built their own adaptive learning system called AMDPC (Adaptation with Multi-Dimensional Personalization Criteria) which can generate learning content to adapt students according their learning styles and cognitive styles. The study showed some significant improvement of the piloting students in both achievement and motivation.

LI (2015) integrated a collaborative technology, Wiki, into learning within a course in teacher education. Along with the face-to-face interactions in the class, students

can use the navigation bar of a wiki site to access various instructions or resources and work on collaborative tasks. The author used a questionnaire to examine the perceived usefulness and attitudes of the participants to the collaborative tool and the effect of the two factors on the different learning style preferences of students.

El-Bishouty et al. (2018) built an adaptive system to support the student in providing course material based on their learning style. They built several lesson resources such as exercises, visual presentations or animations and assessment quizzes and tests for the learners. They proved the effectiveness of the course analyzer can improve the learning of the students with those specific learning styles.

Applying intelligent algorithms to simulate real-world scenarios with the involvement of different learning styles is still a big challenging problem (Khamparia and Pandey 2019). Current researches show a large improvement in supporting learners in the classroom. However, these systems were not organized systematically by a full model which are included all of the three components as mentioned above. On the other hand, these researches concentrated on models separately without the relationship between them. Last but not least, many previous researches have emphasized on student-centric or teacher-centered approach but our model focus on both teaching and learning process and the interactions between all of components in an adaptive learning system.

In an Intelligent Tutoring System, the learning student model takes a very important role and is a decisive factor for the success of an adaptive system for education. In this research, we proposed a novel responsive student model by combining learning style, students' performance, prerequisite relationships, and student real-time responsiveness (by classifying student interesting level in the classroom, based on action recognition solutions used Convolutional Neural Network) to solve adaptation matter in ITS. Our model integrated all the components as the model of Nwana (1990) for a fully ITS. The experiments then show some improvements in performance with several groups of students, comparing to the above papers.

3 Method

3.1 Learning style

Learning styles are patterns that provide direction to learning and teaching. Learning style can be defined as a set of factors, behaviors, attitudes, characteristics that facilitate learning for an individual in a given situation. Learning styles are a popular concept in psychology and are intended to identify how people learn best. Recently, they are chosen in many interactive systems for the teaching and learning process. Among many learning style models such as Myers-Briggs Bishop-Clark and Wheeler (1994), Honey and Mumford (1986), Kolb (1984), and VARK (2020), Felder–Silverman Learning Styles Model (FSLSM) (Felder and Silverman 1988) is the most popular model for adaptive learning. According to Zywno (2003) and Truong (2015), this model is reliable and occupy 70.6% in adaptive learning systems, much greater than Honey & Mumford and Kolb model which were used totally 7.8% in the same systems.

The FSLSM divides learning styles into four dimensions and each dimension follows two opposite directions: active/reflective, sensing/intuitive, visual/verbal, and sequential/global (Fig. 2).

- **Active/Reflective:** This dimension identifies the way you process information. Active learners like to do experiments, interact with objects, doing somethings to study, they like discuss and explain the fact to others. They want to solve problems by groups (Felder and Soloman 1993). Reflective learners do nothing physical, they like to thing and analyze to absorb knowledge. They want to solve by themselves without any support from others. They like to watch and listen to the lecture and think about possible questions (Felder and Silverman 1988).
- **Sensing/Intuitive:** This dimension determines how the learners take in or perceive information. Sensing learners focus on detailed information, they learn by facts and memorize facts very well, they also like doing hands-on (laboratory) work (Felder and Silverman 1988). On the other hand, Intuitive learners are fond of theories, concepts and creations. They hope to dig out the meaning and are more innovative than sensing learners. They like abstract concept, theoretical problems, principles, and mathematical formulations (Felder and Soloman 1993). The intuitive learners like to do the assignment while the sensing learner wants to do a quiz.
- **Visual/Verbal:** This dimension affects the way learners present information. Visual learners like images, graphs and diagrams. They have good ability at remembering what they have seen. On the contrary, Verbal learners want to listen to or read information. They process information and find a solution by verbal. Video is harmonious for Visual learners. Verbal learners want to follow text lessons. They like written or spoken words (Felder and Silverman 1988).
- **Sequential/Global:** This dimension identifies how you process to understand information. Sequential learner like information is expressed step by step. They organize the detailed events to show the whole picture following an order. Global learners tend to approach a problem systematically and globally. They firstly want to see the global picture and then fill with details (Felder and Soloman 1993). Sequential learners prefer to study by text than visual material, so they

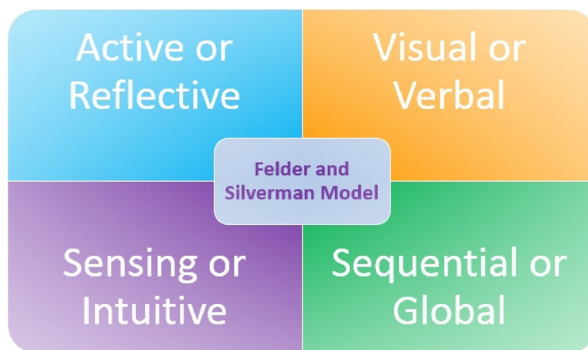


Fig. 2 Felder and Silverman model from Website (2020)

like text or assignment to experience knowledge. However, Global learners want to see the summary images, so they prefer video to the other sources.

To identify learning styles, students must do a Felder-Silverman survey by finishing the Index of Learning Styles (ILS) questionnaire (M.Felder 2020) to evaluate learning styles before going to next steps. The questionnaire has 44 questions with two answers. For each dimension, the result gives a value from +11 to -11 which indicates the preference on the respective dimension. For example, a student is considered an active learner if he/she has an Active dimension with a value higher than 7.

3.2 Learning model

We define a Concept is an elementary unit of knowledge, in the sense that it can not be split into smaller parts. In another word, Concept is the smallest content or the basic of content. To present a Concept, we use the letter C .

A Topic is a pair of (C, W) where C is a set of concepts, $C = \{C_1, \dots, C_n\}$, $W = \{W_1, \dots, W_n\}$ is a set of weights corresponds to each concept C .

A Subject S is a pair of (T, W) where T is a set of topics, $T = \{T_1, \dots, T_n\}$, $W = \{W_1, \dots, W_n\}$ is a set of weights corresponds to each topic T .

To present a Subject we have:

$$S = (T_i^j, W_i^j) \quad (1)$$

To identify a topic T_i which is completed by a student, we use a number called threshold θ . The complete function has a bigger value than this threshold. In our system, we set this number equal to 0.8 (Complete a topic more than 80 percent).

We define a function $Complete(T_i)$ and $Prerequisite(T_i)$ is a possibility of finishing and passing prerequisite the topic, respectively:

$$Complete(T_i) = \begin{cases} 1 & \text{if } \sum_j W_i^j > \theta, \\ 0 & \text{otherwise.} \end{cases} \quad (2)$$

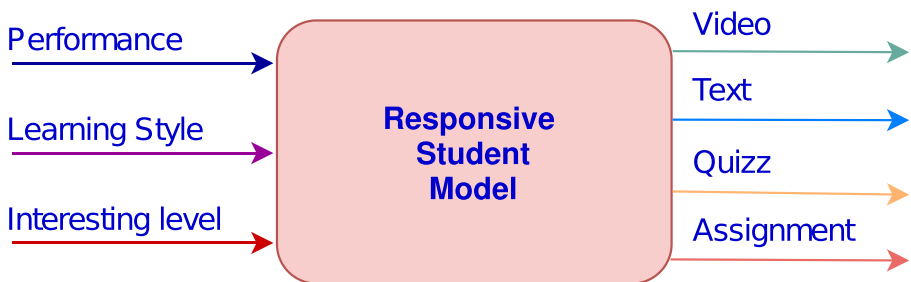


Fig. 3 Responsive student model and the input and output parameters

where W_i^j is weight value of concepts formed by topic T_j

$$Prerequisite(T_i) = \prod_{\substack{j \text{ in set of} \\ \text{prerequisite of } T_i}} Complete(T_j) \quad (3)$$

where T_j is prerequisite of T_i .

3.3 Responsive student model

The responsive student model provides information about student's characteristics or states, such as knowledge, motivation, and attitudes. In our model, we focus on the student model which includes performance, learning style and interesting level (concentration level) to build up a set of parameters for the input of the model. We visualize our model as Fig. 3.

The input of the model includes:

- **Student's Performance:** The performance of the learner in the course. The initial value is 0 for the beginner. In the learning progress, this number is re-evaluated after each lesson.
- **Learning style:** Learning style is identified by doing the ILS questionnaire. All the dimensions are recorded as the user profiles.
- **Interesting level:** This value represents the ability of concentration on the lesson in learning time.

The output of the model includes:

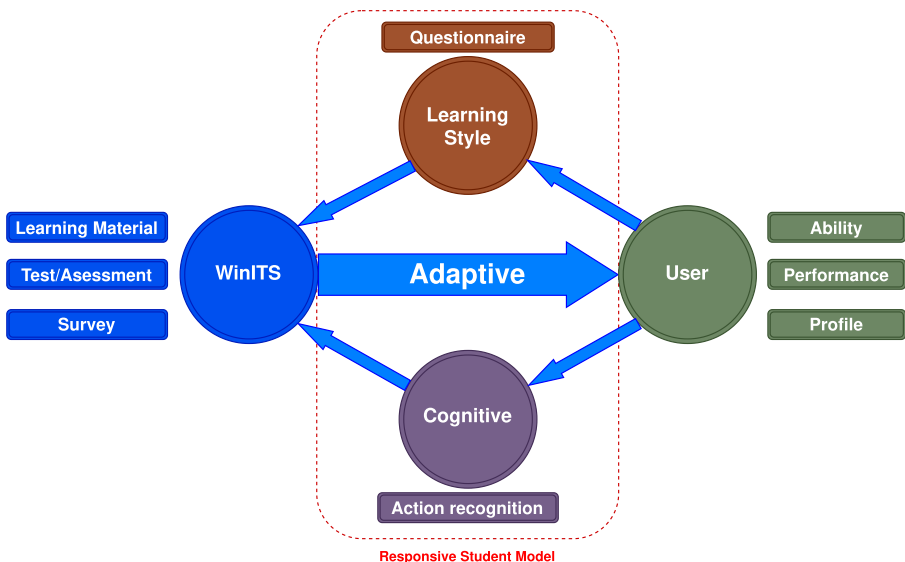


Fig. 4 The relationship between components in WinITS

- **Video:** Video materials are easily accessible to learners of all levels and ages. Learners easily focus on lessons, arouse curiosity in learners. In addition, video materials easily illustrate examples and demonstrate to help learners understand more. The expressions in the videos also make learners more excited, creating motivation for learners to improve their learning ability.
- **Text:** Text materials easy for learners to find, sort and classify. The most popular kind of material ever and traditional. Learners with high concentration can effectively exploit knowledge from this resource.
- **Quiz:** Students easily identify knowledge they know or don't know by doing the quiz. It increases the student-teacher interaction (or automated teaching systems) and increases the learner's ability to focus on the lesson. In addition, it helps learners reorient and adjust the learning content in order to meet the process of answering interactive questions (Mcdaniel et al. 2013).
- **Assignment:** Assignments allow learners to synthesize knowledge and improve their presentation and analysis skills. Assignments will help learners to reflect on their lack of knowledge to make plans to compensate for. Through assignments, learners increase their research ability or generalize problems and focus properly on learning content.

4 Experiment and results

Based on the design of the learning model, we have built an Intelligent Tutoring System called WinITS to reach the adaptive goal for students. The overall structure is illustrated as in Fig. 4. Our system includes 4 main modules:

- **Introduction module:** Introduce the learning model and structure of the course.

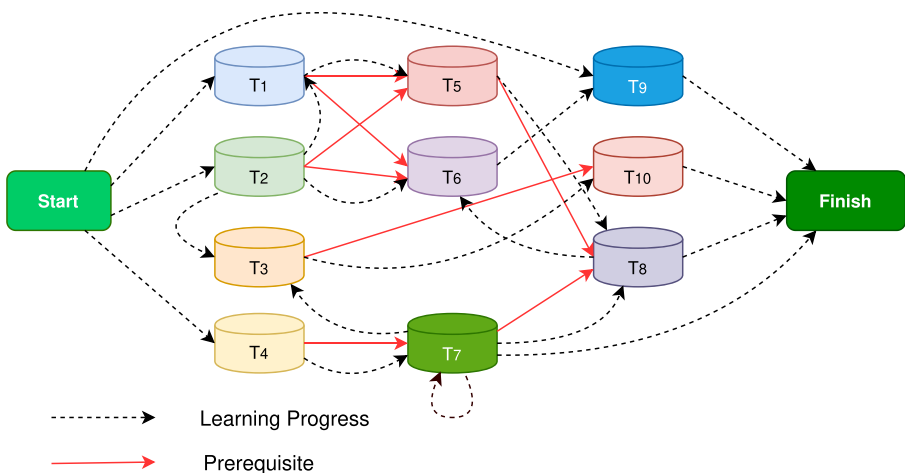


Fig. 5 Learning progress of the experiment lesson. The dot and the solid lines represent for learning progress and prerequisite lessons respectively

- **Learning style module:** The beginning step for the students who launch the system first-time. This phase will evaluate student's learning styles and classify students into each group for estimating material at the next steps.
- **Test module:** The Test module lets the system know the ability of the learner for each stage. This module will calculate function $Complete(T_i)$ for the student.
- **Learning module:** Core module for students to interact with the ITS. This module is integrated with a responsive student model.
- **Survey module:** The built-in questionnaire will collect responses of the learners for analyzing their attitude to the ITS.

The Learning module is a core module based on the user model to identify which sort of material will be used for students to adapt to each student's behavior. According to our researches and experiments, we proposed an adjustable weight table to fit each student's learning style (Table 2). The values in this table were built based on many trial-and-error processes to find out the appropriate parameters. These numbers are also changed in the teaching time by the tutors or administrators.

Each topic has a set of prerequisite conditions that depend on the content of the lessons. A prerequisite topic is content that must be completed prior to another topic. To represent the condition for each topic, we build a relationship among topics which are visualized as in Fig. 5. We use the (3) to calculate the topic's prerequisite by defining the weight for each topic (a pair of (T, W)).

The responsive student model is built base on the mechanism of combining learning style and the status of learners in the classroom. We use four variables SV , ST , SQ , SA to measure the reflection of learning style to the way students study. These variables represent Video, Text, Quiz, Assignment respectively - the four kinds of learning material we have chosen when creating the courseware.

As the work of Binh et al. (2019), we categorize actions in the classroom into 8 groups including "writing", "reading", "raising hand", "looking forward" and "using telephone", "looking up", "looking around", "head down". First of fours represent for *Interesting* status, and the others for *Non-Interesting*. In WinITS, we translate these statuses into "*Interesting Level*" from 1 to 8. To classify these actions, an action recognition module was built to serve the detection process of the student in the classroom. This recognition module is constructed by a Convolutional Neural Network on the Linux-based server, it takes an independent role in identifying a student's concentration ability. A series of student images are sent to the cognitive module to

Table 2 Weight table for students' learning style

Style	Visual	Verbal	Active	Reflective	Sequential	Global	Sensing	Intuitive	Interesting
SV	0.5	0.1	0.4	0.5	0.1	0.4	0.2	0.5	0.1
ST	0.1	0.7	0.1	0.5	0.4	0.1	0.3	0.1	0.5
SQ	0.1	0.3	0.3	0.1	0.3	0.3	0.1	0.2	0.2
SA	0.5	0.1	0.4	0.2	0.1	0.4	0.2	0.4	0.3

get evaluate the interesting level of the students. This value will be updated after an interval of time to let the WinITS decide and re-evaluate variables SA , SQ , ST , SV based on the columns “*Interesting*” in the Table 2. To calculate the value for choosing lesson materials, we calculate the maximum values as the following functions:

$$\max_{\{SV, ST, SQ, SA\}} = \max \sum (v_{LS} * w_{LS} + v_{IL} * w_{IL}) \quad (4)$$

where v_{LS} , w_{LS} , v_{IL} , w_{IL} stand for user’s learning style, weight of learning style, user’s interesting level, interesting weight, respectively.

In the case of extending the selected sort of document, you can easily include your lesson to increase your variable and corresponding weights. This is the flexible characteristic of our ITS to help the teachers build lessons under their circumstances.

4.1 Learning progress algorithm

Student’s learning pathway depends on the level of accumulation of knowledge. Each lesson consists of topics, topics that have not been accumulated will be the next goal in the learning process. An illustrated curriculum can be shown in Fig. 5. In the step **Start**, topics (including concepts) that are indexed in smaller number will be pre-selected (which satisfied prerequisite condition). Here, the topics T_1 , T_2 , T_3 , T_4 , T_9 have no prerequisite, so the learner can choose a starting point, but the concept T_1 is the first selection as the smallest number.

The algorithm of learning progress is shown in Algorithm 1. Based on this algorithm, the initial knowledge about the lesson equals 0 corresponding to learned topic set is empty. Learned topics are accumulated after finishing a topic. The algorithm is ended after complete the lesson or achieved knowledge greater or equal the threshold.

Algorithm 1 Learning progress algorithm in WinITS.

Input : Set of Topics $T = \{T_1, T_2, \dots, T_n\}$,
 A_{req} : Knowledge for accomplishing the course

Output: Learning Process

```

1  $L = \{\}$  //  $L$ : Set of learned Topics
2  $A = 0$  //  $A$ : Accumulated Knowledge
3 while  $T \neq \emptyset$  do
4   if  $Prerequisite(T_i)$  then
5      $L = L \cup T_i$ ;
6      $T = remove(T_i)$ ;
7      $A = A + \alpha_i \tau_i$ ;
8   end
9   if  $A \geq A_{req}$  then
10     $Exit$ ;
11  end
12 end
```

4.2 Leaning object selection algorithm

We propose an algorithm to build the curriculum (Algorithm 2) based on the rules and the weights table (Table 2). The collected data after doing the learning style questionnaire is shown in Table 4 will be calculated the values in the Table 2 by the Algorithm 2. This algorithm is the representation of (4).

Algorithm 2 Learning objects selection based on learner preferences.

Input : Students' parameters
Output: Learning objects

```

/* Initialize */
1  $v_V \leftarrow$  Initialize variable Video selection
2  $v_T \leftarrow$  Initialize variable Text selection
3  $v_Q \leftarrow$  Initialize variable Quiz selection
4  $v_A \leftarrow$  Initialize variable Assignment selection
5 for parameter in Set of parameters do
6    $v_V \leftarrow v_V + SV$ 
7    $v_T \leftarrow v_T + ST$ 
8    $v_Q \leftarrow v_Q + SQ$ 
9    $v_A \leftarrow v_A + SA$ 
   // get values  $SV, ST, SQ, SA$  from Table 2
10 end
11  $v_{max} = MAX(v_V, v_T, v_Q, v_A)$ 
12 if  $v_{max} = v_V$  then select  $Ti_V$ ;
13 if  $v_{max} = v_T$  then select  $Ti_T$ ;
14 if  $v_{max} = v_Q$  then select  $Ti_Q$ ;
15 if  $v_{max} = v_A$  then select  $Ti_A$ ;

```

To figure out the value of SV, ST, SQ, SA , we perform the normalization of the values of learning style in the $[0, 1]$ domain. The maximum value of one learning style dimension is 11, so we divide these values by 11. These values are then multiplied by the number in Table 2 in order to get the values for the variables SV, ST, SQ, SA .

4.3 Student selection process

We applied the designed lessons for all three classes at Hanoi National University of Education in the second term of the 2019-2020 school year. These students were registered into the credit-based system which the subject “The fundamental of Informatics” from the beginning of the semester. The course was selected for the pilot study because of it is usually registered by the first and the second year students, who are new to e-learning systems, this helps us easier to conduct the survey. The three classes consist of 21, 22, 20 students respectively. Two students in class 3 who cannot be involved in the experiment due to coming late and cannot finish the lesson, so we eliminate them from calculations. Hence, the total number of students join into this experiment is 61. They are divided into 2 groups, the Group 1 include class 1 and

2 with 43 students, the Group 2 is all of the students in class 3. We separated these students into 2 groups to compare the effectiveness of the class after piloting.

- **Group 1:** Include 43 students from 2 credit-based classes. Group 1 will study the lesson with the support of WinITS.
- **Group 2:** Consist of 18 students of 1 credit-based class. This group will learn as the traditional class with the same content of lesson as Group 1.

We choose non-probability sampling method because of the natural of distribution of student in a semester when students register themselves. Before conducting the survey, we make a statistic of participating students to ensure the sampling method is suitable. The group dividing process is completely randomized to ensure the objectivity of the experiment. Students in both groups are in the first and the second year and haven't got much prior e-learning experience. The balance of age is similar in both groups. Statistics in the Table 3 show a relatively even distribution between two groups in the test. This guarantees that these groups of students do not have a bias in the distribution of the majors that affect the learning results for the experiment on the subject, which is considered a natural science. The balance in gender between 2 groups illustrate that the procedure for selecting student is fair. The rate is approximately 78%, comparing to 76% of gender ratio in pedagogical schools in Viet Nam, according to the report of Ministry of Education and Training (MOET 2020).

With Group 1, all students go through these steps:

1. Surpass introduction to understand the lesson structure.
2. Do the Felder-Silverman learning style questionnaire.
3. Do the check-point test or skip to the next step.
4. Study lessons or go back the step 3).
5. Do the final test.
6. Do the ITS survey.

With Group 2, they only have to do step 5. The lesson's materials were sent to them for self-studying, at home or school.

Then, we make the assessment by allowing both groups to do the final test to evaluate the performance between two after finishing the designed lesson.

We take some sample photos when Group 1 is learning (Fig. 6. Each computer has an attached camera to capture student's images for action recognition. The user interface of the main screen of WinITS is illustrated as Fig. 7.

Table 3 The distribution of students between two groups

	Group 1		Group 2	
	Count	Percentage	Count	Percentage
Natural Science	13	31.7	6	33.3
Social Science	28	68.3	12	66.7
Male Students	9	22	4	22.2
Female Students	32	78	14	77.8



Fig. 6 In the classroom using WinITS

4.4 Results

In the first phase, the Felder-Silverman learning styles questionnaire was given to 43 students for collecting learning characteristics. The sample data of the survey is shown in the Table 4.

After finishing the survey, data is summarized to compare between dimensions, the results are shown in Table 5.

The results show that students are relatively balanced in Active/Reflective and Sequential/Global dimensions, However, the most significant difference is in the

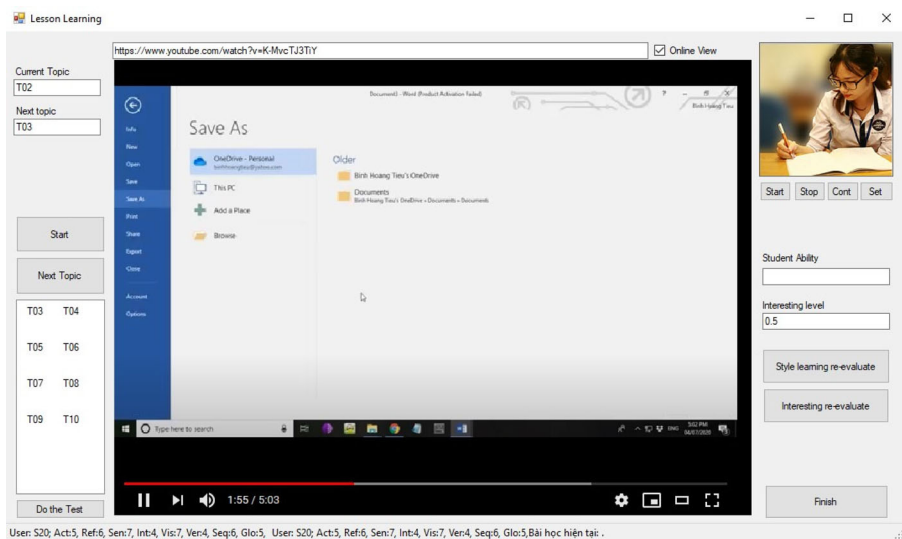


Fig. 7 Main WinITS interface. An integrated camera will track the student's engagement

Table 4 Survey table data structure

StudentID	Active	Reflective	Sensing	Intuitive	Visual	Verbal	Sequential	Global
685611088	8	3	6	5	9	2	5	6
685609016	7	4	9	2	7	4	4	7
675606050	6	5	5	6	11	0	5	6
685105005	9	2	10	1	11	0	11	0
675111047	6	5	6	5	10	1	8	3
685201042	4	7	9	2	11	0	1	10
655111030	7	4	8	3	6	5	7	4
685111028	5	6	10	1	7	4	6	5
685302007	8	3	10	1	4	7	8	3
685906005	5	6	7	4	7	4	6	5

Sensing/Intuitive and Visual/Verbal dimensions, where 86% students came out to be Sensing/Visual and only 14% were Intuitive/Verbal. It shows that most of the students (86%) tend to study by facts and do experiments.

Table 4 lists the learning style data by dimension extracted from 10 students of Group 1. Data from this table will be calculated using Algorithm 2 based on the weight table (Table 2). The results are shown in Table 6. The maximum value is selected corresponding to the learning object.

Table 7 lists the types of materials proposed by the WinITS for learners. The results showed that 67.44% learners were suggested with visual documentation like videos. This shows a correlation with statements in Felder-Silverman's learning style model.

Then, we have conducted two evaluations. Firstly, we compare the examined results between two groups to find the improvement after using the WinITS. Secondly, we conducted a survey for understanding the effect of the software on the first group of students.

The final test included 20 test questions that cover every topic in the lesson. We let all students finish the test without any limitation of the test time.

Table 5 Results of each dimension from the Questionnaire based on Felder-Silverman Learning Styles

Learning Style	Number of Students	Percentage of Students
Active	26	60.5%
Reflective	17	39.5%
Sensing	37	86.0%
Intuitive	6	14.0%
Visual	37	86.0%
Verbal	6	14.0%
Sequential	23	53.5%
Global	20	46.5%

Table 6 Learning object selection variables

Sinh viên	v_V	v_T	v_Q	v_A	Max
685611088	14.3	13.6	11.7	14.2	v_V
685609016	13.1	13.6	14.6	14.8	v_A
675606050	14.9	13	12.9	14.8	v_V
685105005	16.5	11	14.1	13.4	v_V
675111047	14.9	11.8	15.7	15.4	v_Q
685201042	15.6	12.3	12.5	14.3	v_V
655111030	15.1	12.8	12.2	13.8	v_V
685111028	16.4	11.5	13.8	16.3	v_V
685302007	16.3	11.2	14.2	16.8	v_A
685906005	14.3	13.2	13	14.8	v_A

After finishing the final test, we check the test score for both groups. The results are shown in the Table 8 and represented in Fig. 8.

We realized that Group 2 uses less time than Group 1, this can be explained that Group 2 is better prepared because they had received the lesson material before (the teacher sent over the e-mail a day ago). However, the test result of Group 1 is better instead of the average performance score of Group 2 is a little higher than Group 1 (6.6 compared to 6.8). The test score is increased 1.13, compares to 0.79 standard deviation improvement test score from 10 studies (VanLehn 2011).

An interesting number is the average test time of the two groups, Group 1 took the test longer and gained better results, this can be explained that Group 1 tend to do the test more carefully than Group 2.

Furthermore, we group students into three groups based on their GPA. We divide them into 3 groups Under 6.0, From 6.0 to 7.0, Over 7.0 to compare the learning effect to these groups. The result is shown in Table 9. We realize that better improvement happens to the lower GPA groups when their point increases 2.6. This detection can help us to concentrate on the group of students who are facing up the difficulty in learning.

To evaluate the effect of adaptation of ITS to students, we conduct a survey based on the method of evaluations represented by Mousavinasab et al. (2018), including these factors learner experience and learner performance, we add into this survey to scale: the possibility of deployment in teaching and learning. The concrete questions are described in Table 13.

Table 7 Learning objects selection of students

Learning Object	Student Count	Student Rate %
Assignment	10	23.26
Quiz	1	2.33
Text	3	6.98
Video	29	67.44
Total	43	

Table 8 Test score between two groups

		Group 1	Group 2
	Average Studying time	43 minutes	Pre-prepared
	Average Test time	21'31"	16'18"
The Grade Point Average (GPA) are extracted from HNUE credit-based system	Grade Point Average	6.6	6.8
	Average test score	8.4	7.27

First, we summarize the answers by the descriptive statistic as Table 10. Standard deviation measures the variability of the statistical value. It shows the difference in the value of each question or student compared with the mean value. In this table, we can see the standard deviation of both dimensions “Question” and “Student”. As the survey result, the standard deviation of the answers for question and students are 0.12 and 0.17 respectively, this shows that the differences of the answers for each question and student are not too large. We use twelve questions to survey the opinions of the learners in Group 1 who interact with the ITS. These 12 questions are represented in Table 13. In this table we combine the answer into 3 groups: Over Normal: include 4 and 5 points; Normal: 3 points; Under Normal: under 3 to identify the attitude of students. The result of the survey is shown in Table 12. We use the Likert scale which has 5 point levels depending on the student’s answers with 5 responses: Very good, good, normal, bad, very bad.

To compare the relation between question, we calculate the correlation coefficient. In statistic researches, the correlation coefficient is a statistical index measuring the strength and weakness of the relationship between two variables. In our research,

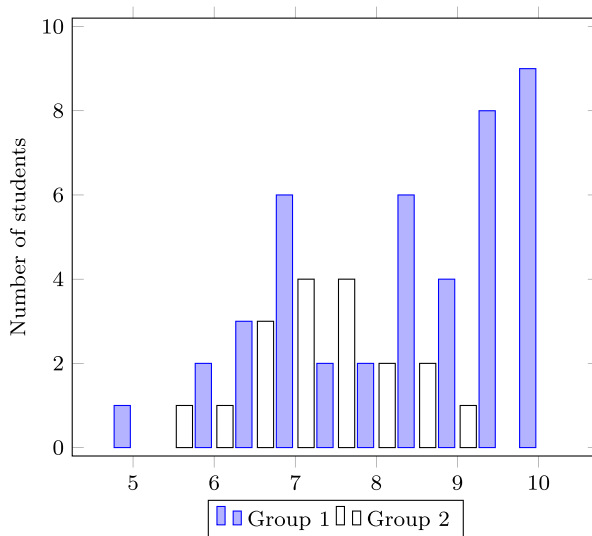


Fig. 8 Test result between two groups, Group 1 uses WinITS and Group 2 doesn't use it. Group 1 shows some better results

Table 9 Analysis with 3 groups according GPA

	Group of Students with GPA	Test score Average	GPA	Difference
The result shows the impact of ITS to the lower groups	Under 6.0	7.89	5.29	2.60
	From 6.0 to 7.0	8.25	6.53	1.72
	Over 7.0	8.94	7.56	1.38

we calculate the Pearson correlation coefficient which is obtained by dividing the covariance of the two variables by the product of their standard deviations.

The results of the correlation analysis are listed in Table 11.

We consider the reliability of the survey result by calculating Cronbach's alpha which is a measure of internal consistency. It is used to measure the scale reliability of the question which are grouped together within an instrument to form scales. Coefficient alpha is the most efficient measure of reliability in terms of the classical model of error measurement and in ideal circumstances (Kline 2015).

Applying this equation to calculate Cronbach's alpha coefficient over 12 items, we have got the value of Cronbach Alpha is 0.8528, the number shows that the items have relatively high internal consistency (if the number is higher than 0.70 is considered "acceptable" in social researches (McCrae et al. 2010)).

According to Table 12, the question **q6** (*According to you, is this learning method more flexible and diverse than traditional learning?*) received the highest point (4.12/5) than the others. In the other hand, the questions **q1** and **q2** (Table 13) received the lowest point (3.53/5). This illustrates that the users feel strange with kind of this application and are not easy to approach. However, they think positive when studying the lesson in this way.

5 Discussion

A comparison with other adaptive learning systems is not similar because of the difference in sample and data. However, we still conduct some collations to other results as a reference. According to our results which are shown in Table 8, the improvement is 15.5% comparing to learning gain of 12% in research of Latham et al. (2014) with

Table 10 Descriptive statistic of the survey of the students who joint into the ITS

	Question	Student
The summary is along to "Questions" and "Students" dimension	Mean	3.89
	Standard Error	0.12
	Median	3.92
	Standard Deviation	0.77
	Max point	4.12
	Min point	3.53

Table 11 Response correlation of the survey

	q1	q2	q3	q4	q5	q6	q7	q8	q9	q10	q11	q12
q1	1.00											
q2	0.25	1.00										
q3	(0.03)	(0.15)	1.00									
q4	0.14	(0.17)	0.64	1.00								
q5	(0.04)	(0.20)	0.55	0.63	1.00							
q6	0.01	(0.05)	0.32	0.49	0.59	1.00						
q7	0.23	0.17	0.37	0.35	0.40	0.42	1.00					
q8	0.08	0.25	0.36	0.38	0.52	0.55	0.74	1.00				
q9	0.24	0.21	0.59	0.54	0.56	0.36	0.53	0.63	1.00			
q10	0.06	0.33	0.41	0.26	0.44	0.39	0.44	0.64	0.67	1.00		
q11	0.31	0.20	0.16	0.34	0.38	0.14	0.26	0.38	0.45	0.44	1.00	
q12	0.02	0.02	0.41	0.47	0.47	0.40	0.48	0.63	0.65	0.44	0.38	1.00

the Adapt group. This research illustration that the improvement of students are not only performance but also motivation as results of Yang et al. (2013).

Adaptive learning has been appointed as a challenging matter of computer science in education. In most previous studies, only learning styles or learning states are considered separately. In this work, we have presented a novel approach by integrating learning styles, learning model and student's responses to adapt students to learning progress and learning materials for a better achievement. We have made an effort to figure out the relationship between personal characteristics and learning objects and incorporate cognitive techniques to identify learner states in the classroom. The proposed model has been tested and validated by using students from real credit-based classes. The obtained results are very promising and show the ability in deploying in schools.

With our achievement, it can be believed that Intelligent Tutorial System with a good learning model can be effectively used along with conventional classroom teaching and as well as a supported tool to help students study by their ways.

To improve the quality of adaptive modules in an ITS, we need more embedded API, machine learning solutions such as emotion recognition, behavior tracking...to make the system smarter, more accurate and convenient.

Table 12 Survey result: the value based on 5-point Likert scale (from 1 to 5) which is similar from not very good to very good

q1	q2	q3	q4	q5	q6	q7	q8	q9	q10	q11	q12
3.53	3.53	4.07	4.07	3.91	4.12	3.77	3.98	3.98	4.07	3.72	3.98

Table 13 Survey results for responses of students to the ITS

ID	Question	Over Normal	Normal	Under Normal
1	Are you familiar and proficient in using an ITS?	48.8	46.5	4.7
2	Do you feel easy when applying intelligent studying methods?	53.5	32.6	14.0
3	Do you think the software will make it easier for you to study?	76.7	23.3	0.0
4	How convenient is this learning method comparing to the traditional learning method?	76.7	20.9	2.3
5	Do you think this ITS will help you achieve better results in your studies	69.8	30.2	0.0
6	According to you, is this learning method more flexible and diverse than traditional learning?	81.4	16.3	2.3
7	Is the lesson organization, teaching/learning model suitable for your current lesson?	62.8	34.9	2.3
8	Do you think this adaptive learning technology will support you better?	79.1	20.9	0.0
9	Are you interested in teaching and learning with this software?	69.8	25.6	4.7
10	Do you think the time will be shortened if applied this learning method?	79.1	18.6	2.3
11	Do you find it easy to create an e-lecture?	58.1	39.5	2.3
12	How acceptable to apply this teaching software to the school?	79.1	18.6	2.3

The unit in percentage

6 Conclusion and future work

It is concluded that the responsive student model developed for the WinITS has made a positive improvement in student learning outcome. Students whose tutorial was adapted to match their learning styles performed on average 15.5% better on final test scores than students who were learned in conventional ways. The students' feedback shows some promise comments and will be a recommendation for adaptive learning systems.

On the other hand, our study shows some limitations. Firstly, the sample size of the experiment was not large, therefore, the findings could not infer to generalization case. Secondly, the experiments have only integrated action recognition to indicate the state of students in the classroom. This work seems not to be enough persuaded. Moreover, the system should be improved by automatically generating the lesson for the students, instead of just choosing available documentaries as the current application, it might reduce the adaptation of the system.

From the responses of the learners, it can be seen that the proposed approach is promising. To deploy this model to other applications, teachers or tutors only need to provide new sorts of materials such as text, video, audio, presentations..., then extend the weight table and append resources for the learners.

In our future research, we plan to conduct the proposed approach with a larger sample size to reach more students and receive more responses. We can extend the student model by including more properties such as Bloom level or emotional states during the lesson to control possible factors that can affect students' results. In the meantime, we need to transfer this system to a web-based format for widening users and touching out many more students.

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